

Name:

INTERNATIONAL GRAMMAR SCHOOL

1999

MATHEMATICS

4 UNIT

HALF YEARLY EXAMINATION

YEAR 12

**Time allowed --- 2 hours
(Plus 5 minutes reading time)**

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Questions are NOT of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a *new page*. Number each question clearly.
- Label each page with your name.

Question1: (20 marks) Start a new page.

[8]

(a) $z_1 = 1 + 3i$, $z_2 = 1 - i$

- i. Find in the form $a + ib$, where a and b are real, the numbers $z_1 z_2$ and $\frac{z_1}{z_2}$.
- ii. On an Argand Diagram the vectors $\vec{OA}$, $\vec{OB}$ represent the complex numbers $z_1 z_2$ and $\frac{z_1}{z_2}$ respectively (where z_1 and z_2 are given above). Show this on an Argand Diagram, giving the coordinates of A and B . From your diagram, deduce that $\frac{z_1}{z_2} - z_1 z_2$ is real.

[6]

- (b) $-3 + 4i$ has two square roots z_1 and z_2 . Find z_1 and z_2 in the form $a + ib$ and show the points representing $-3 + 4i$, z_1 and z_2 on an Argand Diagram. Show that these three points are the vertices of a right angled triangle.

[6]

- (c) The complex number z is represented by the point P on an Argand Diagram. Indicate clearly on a single diagram the locus of P in each of the following cases:

- i. $|z - 4| = |z + 2i|$
- ii. $\arg(z + 3) = \frac{\pi}{4}$

Show that there is a point representing a complex number of the form ib , where b is real, which lies on both loci.

ion2: (15 marks) Start a new page

[6]

- (a) i. Expand $z = (1 + ic)^6$ in powers of c .

- ii. Hence find the five real values of c for which z is real.

[4]

(b)

Let $w = \frac{3 + 4i}{5}$ and $z = \frac{5 + 12i}{13}$, so that $|w| = |z| = 1$.

- i. Find wz and $w\bar{z}$ in the form $x + iy$.
- ii. Hence find two distinct ways of writing 65^2 as the sum $a^2 + b^2$, where a and b are integers and $0 < a < b$

[5]

(c)

- i. Show that $(1 - 2i)^2 = -3 - 4i$
- ii. Hence solve the equation $z^2 - 5z + (7 + i) = 0$.

Question 3: (15 marks) Start a new page

[8]

- (a)i. Sketch the graph of $f(x) = x^3 - 3x$ showing clearly the coordinates of any points of intersection with the x axis and the coordinates of any turning points.
- ii. Use the graph of $y = f(x)$ in part (i) to sketch the graph of $y = |f(x)|$ showing clearly the coordinates of any critical points (where $\frac{dy}{dx}$ is not defined) and the coordinates of any turning points.
- iii. Use the graph of $y = f(x)$ in part (i) to sketch the graph of $y = \frac{1}{f(x)}$ showing clearly the equations of any asymptotes and the coordinates of any turning points.

[7]

- (b)i. On the same set of axes, sketch and label clearly the graphs of the functions $y = x^{\frac{1}{3}}$ and $y = e^x$.
- ii. Hence, on a different set of axes, without using calculus, sketch and label clearly the graph of the function $y = x^{\frac{1}{3}}e^x$.
- iii. Use your sketch to determine for which values of m the equation $x^{\frac{1}{3}}e^x = mx + 1$ has exactly one solution.

Question 4: (15 marks) Start a new page

[10]

Let $f(x) = -x^2 + 6x - 8$. On separate diagrams, and without using calculus, sketch the following graphs. Indicate clearly any asymptotes and intercepts with the axes.

- i. $y = f(x)$
- ii. $y = |f(x)|$
- iii. $y^2 = f(x)$
- iv. $y = \frac{1}{f(x)}$
- v. $y = e^{f(x)}$

[5]

- i. If α is a double zero of the polynomial $P(x)$, show that α is a zero of $P'(x)$.
- ii. $(x - 1)^2$ is a factor of $x^5 + 2x^4 + ax^3 + bx^2$. Find the values of a and b.

Question 5: (15 marks) Start a new page

[3]

i. Express $\sqrt{3} - i$ in modulus-argument form.

ii. Hence evaluate $(\sqrt{3} - i)^8$

[5]

$\sqrt{3} + i$ is one root of $x^4 + px^2 + q = 0$, where p and q are real. Find p and q and factor $x^4 + px^2 + q$ into quadratic factors with real coefficients.

[7]

The quadratic equation $x^2 - x + k = 0$, where k is a real number, has two distinct positive real roots α and β .

i. Show that $0 < k < \frac{1}{4}$.

ii. Show that $\alpha^2 + \beta^2 = 1 - 2k$ and deduce that $\alpha^2 + \beta^2 > \frac{1}{2}$.

iii. Show that $\frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8$

THE END

4 UNIT ANSWERS

Question 1.

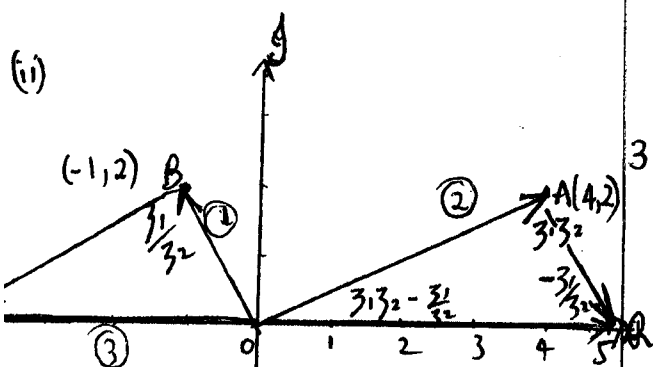
$$\begin{aligned} (i) z_1 z_2 &= (1+3i)(1-i) \\ &= 1+3+i(-1+3) \\ &= 4+2i \end{aligned}$$

$$\frac{z_1}{z_2} = \frac{(1+3i)}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{1-3+i(1+3)}{1+1}$$

$$= \frac{-2+4i}{2}$$

$$= -1+2i$$



3 see diagram

$$\begin{aligned} z_1 z_2 + \frac{z_1}{z_2} &= (-4-1) + i(2-2) \\ &= -5 \\ &= \text{Real} \end{aligned}$$

$$\begin{aligned} i) -3+4i &= (a+ib)^2 \\ -3+4i &= a^2+2aib-b^2 \end{aligned}$$

equating R and I parts.

$$a^2-b^2 = -3 \dots \dots \textcircled{1}$$

$$2ab = 4 \dots \dots \textcircled{2} \Rightarrow b = \frac{2}{a}$$

substitution ② into ①

$$a^4 + 3a^2 - 4 = 0$$

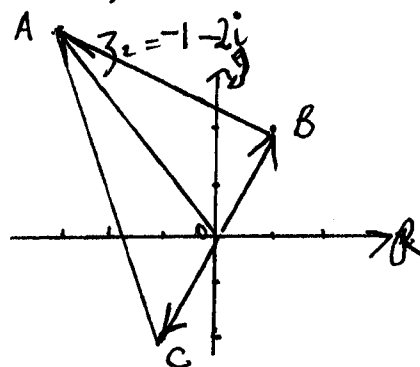
$$(a^2-1)(a^2+4) = 0$$

$$a^2 = 1 \text{ or } (a^2 = 4 \text{ not possible})$$

as a is real.

$$a = \pm 1 \therefore b = \pm 2$$

$$\therefore z_1 = 1+2i$$



let OA represent $-3+4i$

OB represent $1+2i$

OC represent $-1-2i$

$$\therefore AB \text{ is } -3+4i - (1+2i) = -4+2i$$

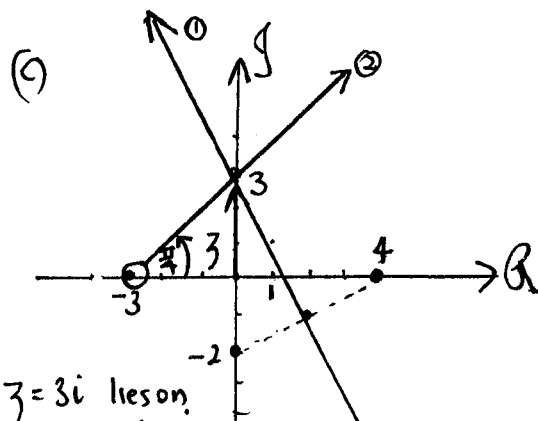
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$$\begin{aligned} \text{and } OB \times 2i &= (1+2i)2i \\ &= -4+2i \end{aligned}$$

So $AB \perp OB$ and

OB and OC are collinear

So ABC is a rt $\angle \Delta$



2 idcus

$z = 3i$ lies on

2

Question 2:

(i) $z = (1+ic)^6$

using binomial expansion

$$z = 1 + \binom{6}{1}ic + \binom{6}{2}(ic)^2 + \binom{6}{3}(ic)^3 + \dots + \binom{6}{6}(ic)^6$$

$$z = 1 - 15c^2 + 15c^4 - c^6 + i(6c - 20c^3 + 6c^5)$$

z is real when $6c - 20c^3 + 6c^5 = 0$

$$2c(3 - 10c^2 + 3c^4) = 0$$

$$2c(3c^2 - 1)(c^2 - 3) = 0$$

$$\therefore c = 0, c = \pm \frac{1}{\sqrt{3}}, c = \pm \sqrt{3}$$

2
(-1/2 each error)

b) $w \cdot z = \left(\frac{3+4i}{5}\right) \times \left(\frac{5+2i}{13}\right)$
 $= \frac{1}{65} (15 - 48 + i[36 + 20])$
 $= \frac{1}{65} (-33 + 56i)$

$w \cdot \bar{z} = \left(\frac{3+4i}{5}\right) \times \left(\frac{5-2i}{13}\right)$
 $= \frac{1}{65} (15 + 48 + i[-36 + 20])$
 $= \frac{1}{65} (63 - 16i)$

given $|w| = |z| = 1$

$$\therefore |wz| = \sqrt{\left(\frac{1}{65}\right)^2 [33^2 + 56^2]} = 1$$

$$\Rightarrow 33^2 + 56^2 = 65^2$$

$$|w\bar{z}| = \sqrt{\left(\frac{1}{65}\right)^2 [63^2 + 16^2]} = 1$$

$$\Rightarrow 63^2 + 16^2 = 65^2$$

$$16^2 + 63^2 = 65^2$$

(c) $(1-2i)(1-2i) = 1 - 4 + i(-2-2)$
 $= -3 - 4i$ qed

(ii) $z^2 - 5z + (7+i) = 0$

$$z = \frac{5 \pm \sqrt{5^2 - 4(7+i)}}{2}$$

$$= \frac{5 \pm \sqrt{25 - 28 - 4i}}{2}$$

$$= \frac{5 \pm \sqrt{-3 - 4i}}{2}$$

$$= \frac{5+1-2i}{2} \text{ or } \frac{5-1+2i}{2}$$

$$z_1 = 3-i \text{ or } z_2 = 2+i$$

15

Question 3:

(a) $f(x) = x^3 - 3x$

$$f(x) = 0 \Rightarrow x^3 - 3x = 0$$

$$x(x^2 - 3) = 0$$

$$x = 0, x = \sqrt{3}, x = -\sqrt{3}$$

$$f'(x) = 3x^2 - 3$$

$$f'(x) = 0 \Rightarrow x^2 - 1 = 0$$

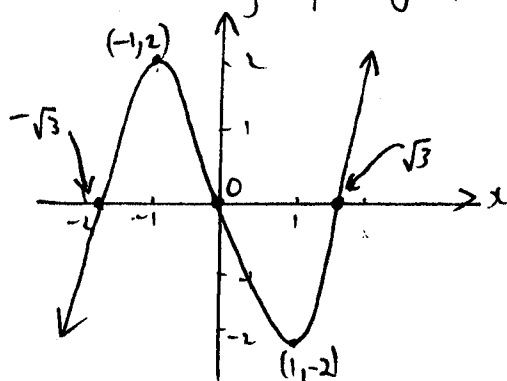
$$\Rightarrow x = 1, x = -1$$

$$f(1) = -2, f(-1) = 2$$

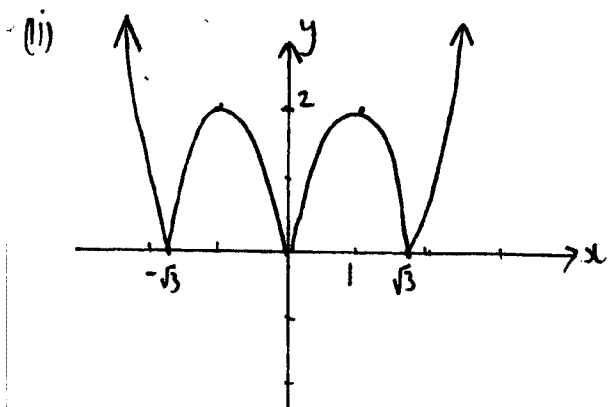
$$(1, -2) \quad (-1, 2)$$

$\min f''(1) > 0 \quad \max f''(-1) < 0$

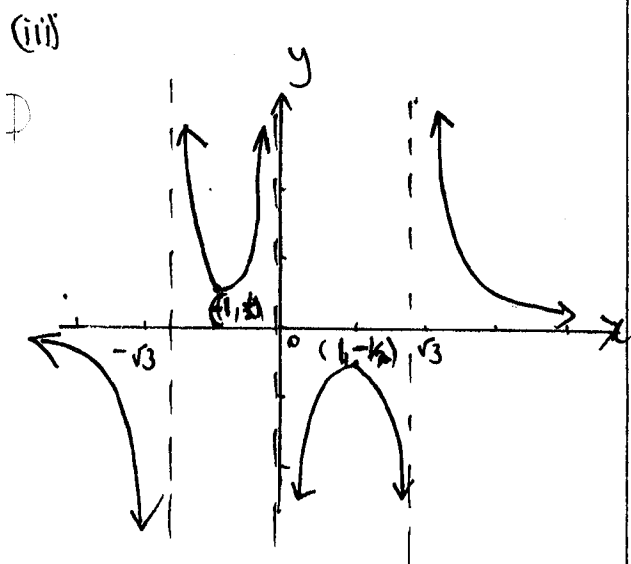
$$f''(x) = 6x \Rightarrow x = 0 \text{ is a possible point of inflexion}$$



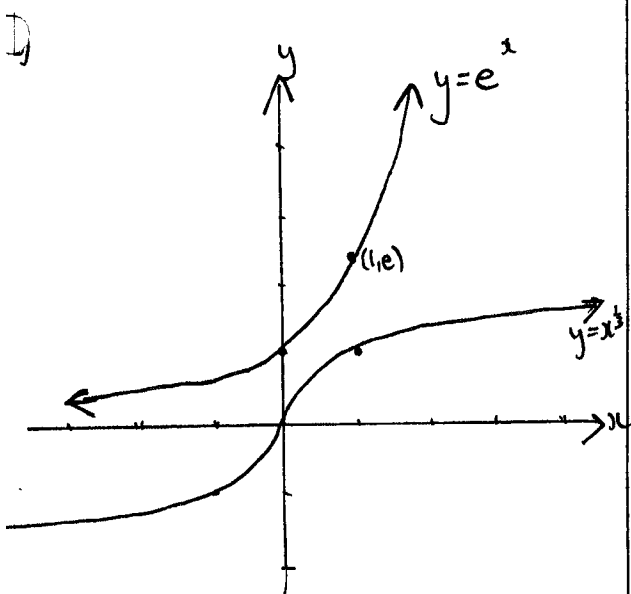
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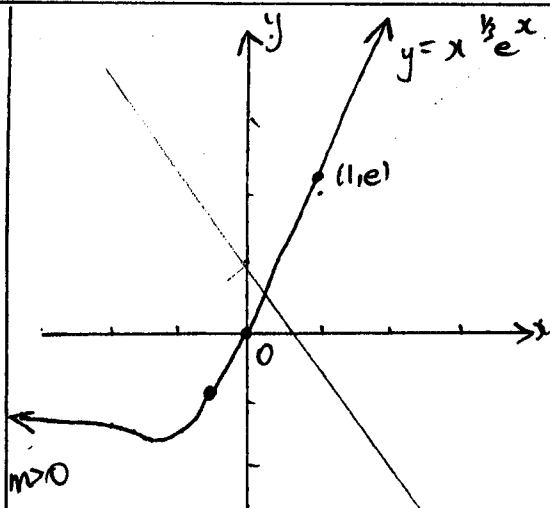
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2



3



2

(iii) $x^{1/3}e^x = mx + 1$

let $y = mx + 1$ (passes through $(0, 1)$)

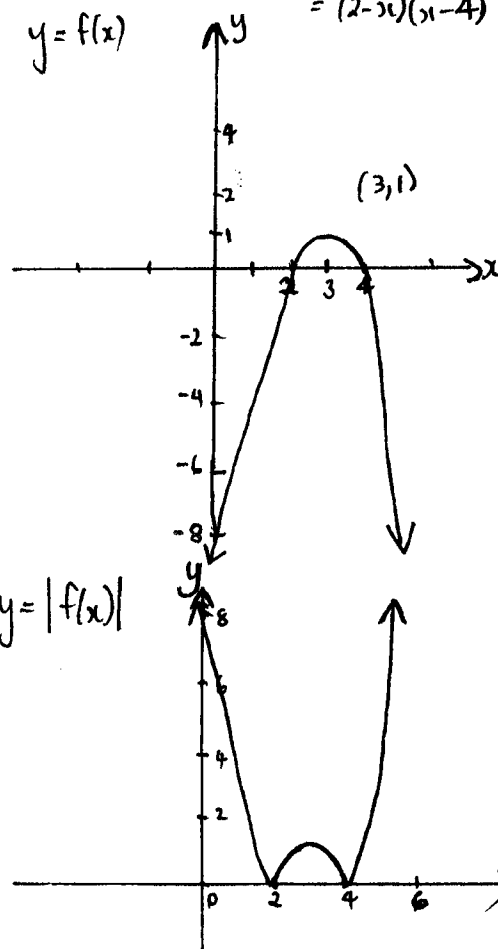
see diagram
 $\therefore$ exactly one solution for $m \leq 0$

1 line

1 ans

15

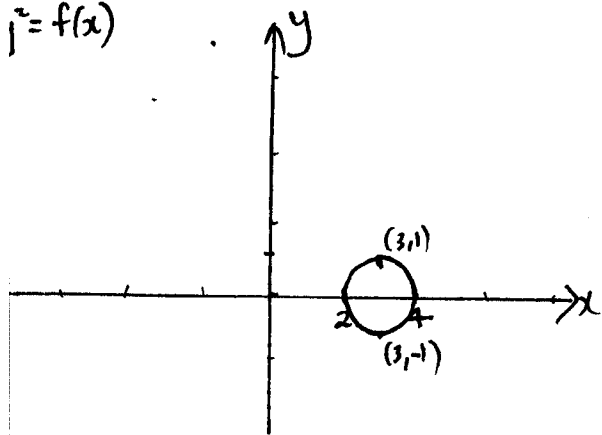
Question 4: $f(x) = -x^2 + 6x - 8$
 $= (2-x)(x-4)$
 $y = f(x)$



2

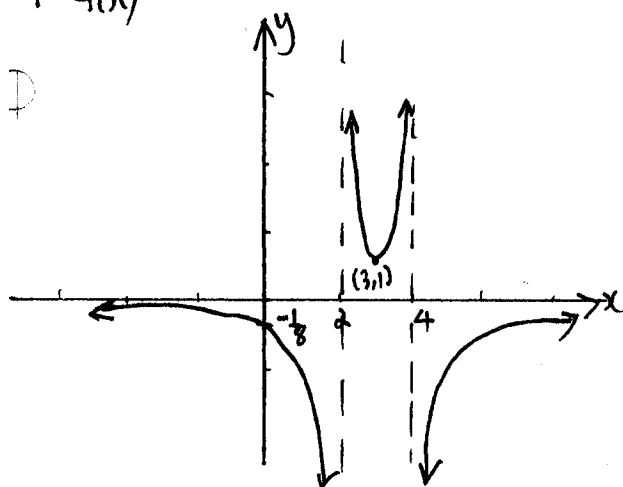
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$$y = f(x)$$



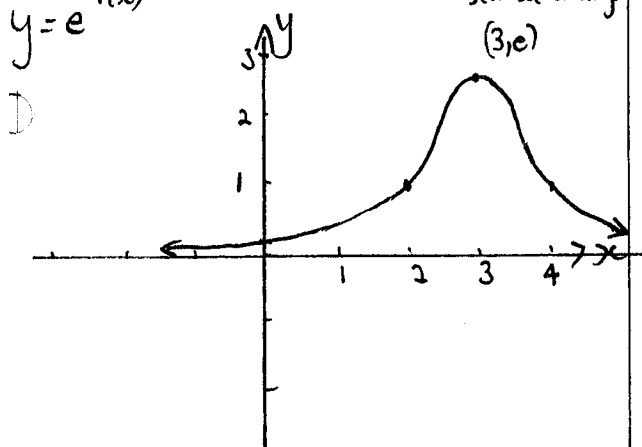
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$$y = \frac{1}{f(x)}$$



2

$$y = e^{f(x)}$$



2

$$y = e^{f(x)}$$

$$y' = f'(x) \cdot f(x) e^{f(x)}$$

Note: $e^0 = 1$ when $x = 2, 4$

$$f(x) = -x^2 + 6x - 8$$

$$f'(x) = -2x + 6$$

$$f'(x) = 0 \text{ when } x = 3$$

$$f(x) = 0 \text{ when } x = 2 \text{ or } x = 4$$

$$(b) i) P(x) = (x - \alpha)^2 Q(x) \text{ for some } Q(x)$$

$$P'(x) = 2(x - \alpha)Q(x) + (x - \alpha)^2 Q'(x)$$

$$= (x - \alpha)[2Q(x) + (x - \alpha)Q'(x)]$$

So α is a zero of $P'(x)$

$$(ii) P(x) = x^5 + 2x^4 + ax^2 + bx^2$$

$$P'(x) = 5x^4 + 8x^3 + 3ax^2 + 2bx$$

$$P(1) = 1 + 2 + a + b = 0$$

$$\Rightarrow a + b = -3 \dots \textcircled{1}$$

$$P'(1) = 5 + 8 + 3a + 2b = 0$$

$$\Rightarrow 3a + 2b = -13 \dots \textcircled{2}$$

$$2 \times \textcircled{1} \Rightarrow 2a + 2b = -6$$

$$\textcircled{2} \Rightarrow 3a + 2b = -13$$

$$\textcircled{1} - \textcircled{2} \Rightarrow -a = +7$$

$$\therefore a = -7$$

$$\therefore b = 4$$

Question 5:

$$\sqrt{3} - i = r \operatorname{cis} \theta$$

$$\text{where } r = \sqrt{3^2 + 1^2} = 2$$

$$\text{and } \tan \theta = \frac{-1}{\sqrt{3}}$$

$$\theta = -\frac{\pi}{6}$$

$$= 2 \operatorname{cis} \left(-\frac{\pi}{6}\right)$$

$$(\sqrt{3} - i)^8 = 2^8 \left(\cos -\frac{\pi}{6} + i \sin -\frac{\pi}{6}\right)^8$$

$$= 2^8 \left(\cos \frac{8\pi}{6} - i \sin \frac{8\pi}{6}\right)$$

$$= 2^8 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2}\right)$$

$$= 2^7 (-1 + i\sqrt{3})$$

If $\sqrt{3} + i$ is a root of

$$x^4 + px^2 + q = 0 \text{ where } p, q \in \mathbb{R}$$

then $\sqrt{3} - i$ is also a root

$$\Rightarrow \text{Quadratic factor } (x - (\sqrt{3} + i))(x - (\sqrt{3} - i))$$

$$= (x^2 - 2\sqrt{3}x + 4)$$

By polynomial division,

$$\begin{array}{r} x^2 + 2\sqrt{3}x + (p+8) \\ x^2 - 2\sqrt{3}x + 4 \end{array}$$

$$\begin{array}{r} x^2 - 2\sqrt{3}x + 4 \overline{) x^4 + 0x^3 + px^2 + 0x + q} \\ x^2 - 2\sqrt{3}x^3 + 4x^2 \end{array}$$

$$\begin{array}{r} x^2 - 2\sqrt{3}x^3 + 4x^2 \overline{) x^4 + 0x^3 + px^2 + 0x + q} \\ 2\sqrt{3}x^3 + (p-4)x^2 + 0x \end{array}$$

$$\begin{array}{r} 2\sqrt{3}x^3 + (p-4)x^2 + 0x \overline{) 2\sqrt{3}x^3 + 12x^2 + 8\sqrt{3}x} \\ (p+8)x^2 - 8\sqrt{3}x + q \end{array}$$

$$\begin{array}{r} (p+8)x^2 - 8\sqrt{3}x + q \overline{) (p+8)x^2 - 2(p+8)\sqrt{3}x + 4(p+8)} \\ 0 \quad 0 \end{array}$$

$$(p+8)x^2 - 2(p+8)\sqrt{3}x + 4(p+8)$$

$$\begin{array}{r} (p+8)x^2 - 2(p+8)\sqrt{3}x + 4(p+8) \overline{) (p+8)x^2 - 2(p+8)\sqrt{3}x + 4(p+8)} \\ 0 \quad 0 \end{array}$$

$$\therefore 2(p+8)\sqrt{3} = 8\sqrt{3}$$

$$p+8 = 4$$

$$p = -4 \therefore q = 4(-4+8)$$

$$q = +16$$

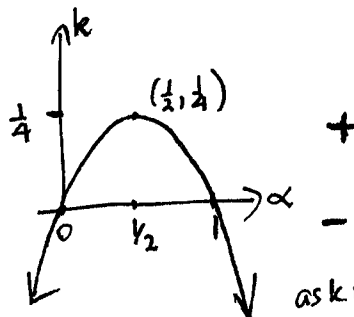
$$c) \text{ let } x^2 - x + k = (x - \alpha)(x - \beta)$$

$$(i) \therefore \alpha + \beta = 1 \dots \textcircled{1}$$

$$\alpha\beta = k \dots \textcircled{2}$$

$$\text{sub } \textcircled{1} \text{ into } \textcircled{2} \Rightarrow$$

$$\alpha(1 - \alpha) = k$$



$$\therefore 0 < k < \frac{1}{4}$$

ask is a positive real

$$\begin{aligned} (ii) \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= 1^2 - 2k \\ &= 1 - 2k \end{aligned}$$

$$\text{but } 0 < k < \frac{1}{4}$$

$$\Rightarrow 0 < 2k < \frac{1}{2}$$

$$-\frac{1}{2} < -2k < 0$$

$$\frac{1}{2} < 1 - 2k < 1$$

$$\frac{1}{2} < \alpha^2 + \beta^2 \quad \text{qed}$$

$$(iii) \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2 \beta^2} = \frac{1 - 2k}{k^2}$$

$$\frac{1 - 2k}{k^2} > \frac{1}{\left(\frac{1}{4}\right)^2}$$

$$\frac{1 - 2k}{k^2} > \frac{16}{2}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8 \quad \text{qed}$$

2

15 min